

Historic Landmarks Insurance: The Art of Prediction

Summary

It is well known that the global climate is warming and that **extreme weather events** are becoming more frequent as a result of increasing activities such as the burning of fossil fuels. Changing meteorological conditions increase the risk of damage to historic landmarks. It is extremely important to construct a model to estimate **the cost of protecting and insuring historical landmarks** in the future, which can help people make a reasonable judgment on whether to underwrite a historical landmark or not, so that historical landmarks can be better protected under the increasingly severe environmental conditions.

For the **first** problem, we can use MATLAB to build a multiple regression model to deal with the data of various meteorological conditions, and adopt methods including Monte Carlo simulation to test the aspects of multiple regression equations. From this, the **relationship between the various weather conditions and the percentage of insured losses** can be derived and the validity of the model can be practically tested at another historical landmark. The resulting percentage of insured loss is then used to **calculate the actual insured loss** and thus **make a judgment** about whether or not a particular landmark is insured against **catastrophe**.

For the **second** problem, we can use the multiple linear regression model obtained in the first question and substitute various local weather conditions to analyze **whether or not property development should be carried out** in a certain location.

For the **third** problem, we can first **predict the future trend** of various meteorological conditions in a certain place by using one-way linear regression. The predicted meteorological data are then brought into the multiple linear regression equation obtained in the first question. Then we take a similar approach as in the first question to get the actual insurance loss, **determine whether it should be insured** against catastrophe, and give corresponding analytical **suggestions for the protection and insurance** of historical landmarks.

Key words: Multiple linear regression, Monte Carlo simulation, one-way linear regression

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I. Introduction

1.1 Problem Background

In recent years, global climate change has led to an increasing number of extreme weather events, which have had an enormous impact on many aspects of human life, including society and economy. Relevant international studies have concluded that the causes of climate change are mainly the alteration of the Earth's ecological balance as a result of higher carbon dioxide concentrations, higher methane concentrations and changes in land use and land cover. Climate change not only produces extreme weather, causing significant human morbidity and mortality, but also poses significant risks to mental health. In addition to this, climate change will also have an impact on the financial industry and the business environment, where extreme weather events have increased the incidence of natural disasters, and there has been a significant rise in catastrophe insurance claims in the property insurance industry.

Catastrophe refers to a natural disaster event that causes particularly large damage and loss to people's lives and property, and has a serious impact on the regional or national economy and society. Catastrophe insurance refers to the diversification of risks through the catastrophe insurance system for risks that may cause huge property losses and serious casualties due to natural disasters such as earthquakes, hurricanes, tsunamis and floods. In order to enhance the efficiency and fairness of government disaster relief, catastrophe insurance is gradually becoming the preferred option for utilizing market-based disaster relief mechanisms.

However, climate deterioration has led to an increase in the frequency of catastrophes, and losses due to catastrophe risk are characterized by huge and concentrated losses, making the insurance policies underwritten by insurance companies excessively risky and leading to a situation where insurance companies are faced with paying out too many claims. As a result, risk premiums have increased, insurers have increased pricing and have also had to change how and where they are willing to insure, and insurers have faced shocks to their sustainability by deciding whether to underwrite operation in areas where extreme weather occurs as well as by making decisions based on forecasts of weather deterioration and building appropriate models to improve insurers' ability to protect against future claims costs.

1.2 Restatement of Problem

Considering the background, we need to solve the following problems:

- Develop a model for insurance companies as a way to determine if they should underwrite in areas with increased extreme weather events.
- Derive and test our model using two historical landmarks.
- Use the insurance model to predict the cost of maintenance for one historic landmark.
- Analyze the model to give appropriate preservation recommendations for that historic landmark.

1.3 Our Work

First, based on the background of this paper and some literature, the temperature, precipitation, and the number of occurrences of natural disasters at the location of the historical landmark Yellowstone, Wyoming, to be measured were statistically determined, and we were asked to build a multivariate linear regression model, apply the model to evaluate the cost of insurance in relation to the climatic conditions and the number of occurrences of extreme weather, and then test the accuracy of this multivariate linear regression with another location, Shenyang Imperial Palace, and then build a model for predicting the weather conditions and the number of occurrences of extreme weather, and then use the model to predict the future climate of the third location, New York Statue of Liberty, and then use the multivariate linear regression that had previously been built to calculate future premiums and to help the insurance company to make a decision.

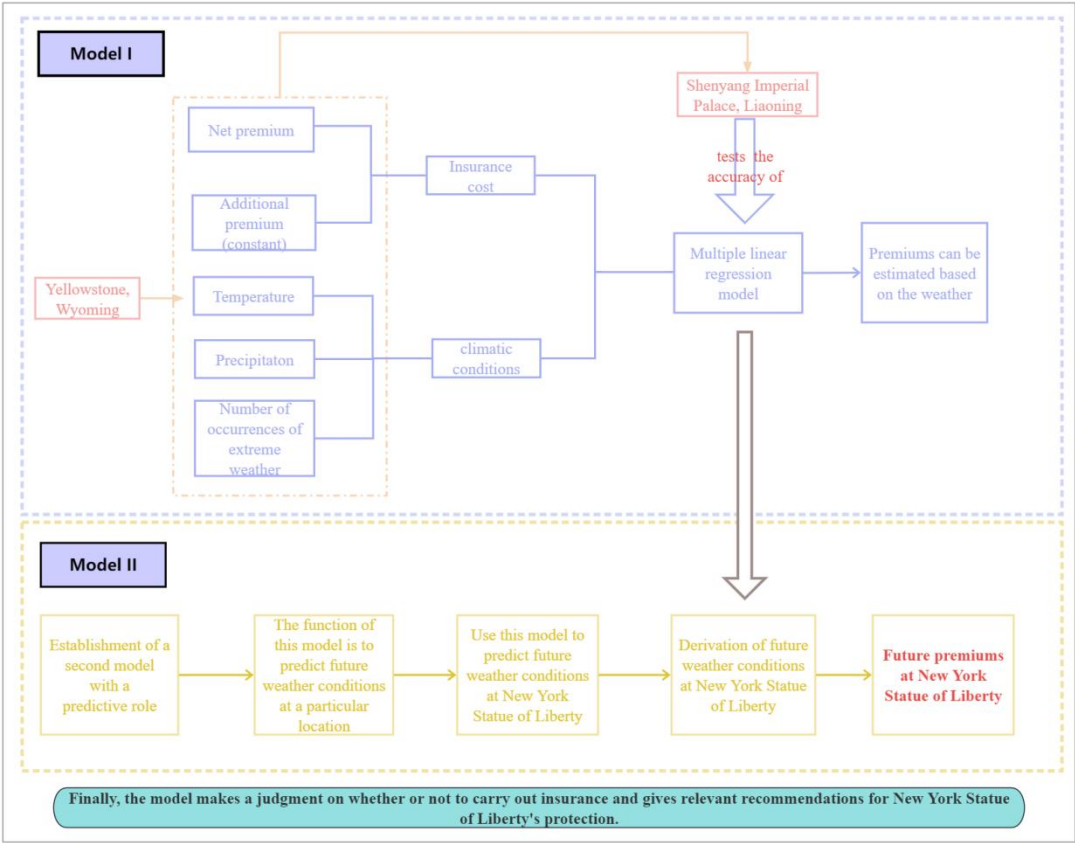


Figure1: Our Work

II. The Description of the Problem

According to the background of the topic and some literature and data, the climatic conditions of Yellowstone Park in Wyoming and the Forbidden City in Shenyang, which are two areas from different continents, are selected to use multiple linear regression to derive and demonstrate a model used to judge whether insurance should be underwritten under what circumstances; the resulting insurance model is

adjusted to determine whether development should be carried out in certain locations. Considering the historical significance of the selected historic landmark and the worldwide concern for its preservation, the Statue of Liberty is finally adopted as the focus of the analysis; finally, the model is applied to analyze it specifically and to obtain the theoretical basis for making preservation recommendations.

Multiple linear regression is one of the important methods to solve this problem. The reasons for choosing this method are as follows: through the multiple linear regression model, we can quickly and intuitively analyze the linear relationship between the amount of insurance payout and the number of precipitation, temperature, and the occurrence of extreme weather. The correlation between the variables is accurately measured and close to the actual situation. It is easy to operate and yields unique results for the same set of data.

III. Models

QUESTION 1

3.1 Basic Model of Question 1

3.1.1 Terms, Definitions and Symbols

In this section, in order to facilitate the use of symbols and the reader's reading in subsequent articles, we will give the use of symbols in the model with their corresponding explanations. Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Terms, definitions and symbols in basic model 1

Symbol	Definitions
$\beta_i(i=1,2,3)$	Regression coefficient of the i-th independent variable
β_0	Error term
T	Temperatures
P	Precipitation
W	Number of occurrences of extreme weather
R^2_{adjusted}	Adjusted correlation coefficient
VIF	Variance inflation factor
$R^2_{1-3/m}$	Using the m-th of the three independent variables as the dependent variable, the goodness of fit of the other independent variables
Y	Amount of insurance claims (cost of insurance)
$\rho_{\beta_0,T}$	Correlation coefficient of error term with temperature
$\rho_{\beta_0,P}$	Correlation coefficient between the error term and precipitation amount
$\rho_{\beta_0,W}$	Correlation coefficient between the error term and the number of extreme weather events

3.1.2 Assumptions

Assumption1: The percentage of insured losses is influenced by three factors: temperature, precipitation, and the number of extreme weather events.

Justification: With global warming, temperatures are gradually increasing, precipitation is trending upwards and the number of extreme weather events is gradually increasing. This leads to more natural erosion of landmarks and a higher probability of natural disasters. Therefore, this paper adopts these three factors as the main factors affecting the percentage of insurance losses.

Assumption2: Precipitation, temperature, and the number of extreme weather occurrences are linearly related to the dependent variable of insurance costs.

Justification: In the paper, the joint significance test and the significance test with β_i were conducted and the adjusted goodness of fit was found to be significant and good, so the variables can be considered as a linear relationship.

Assumption3: The error β_0 is a spherical perturbation term, i.e., it satisfies variance chi-square and is uncorrelated with the three independent variables: precipitation, temperature, and weather extremes.

Justification: The Monte Carlo test for endogeneity and the BP test for heteroskedasticity are performed in the following sections, respectively, and the results show that the endogeneity is not significant and the error term is not significantly correlated with any of the independent variables or their quadratic terms, so that it can be regarded as a spherical perturbation term.

Assumption4: The error terms are normally distributed.

Justification: This error term is influenced by a variety of independent factors and is a random variable, while the distribution of a large number of independent random variables tends to be normal, so the error term should follow a normal distribution.

3.1.3 The Foundation of Model

In establishing the effect of extreme weather and climate change on insurance costs, a multiple linear regression model was constructed using MATLAB to predict the percentage of insurance payouts from local precipitation, temperature, and the number of extreme weather occurrences. The model was tested for heterogeneity, significance of regression coefficients, endogeneity, multicollinearity, and internal variance using MATLAB to ensure the reliability of the model predictions. The percentage of payout was determined to determine whether a historical landmark should be insured against catastrophe and the corresponding premiums to be charged.

Procedures:

Multiple linear regression modeling to determine the impact of extreme weather and climate change on the percentage of insurance claims and to finalize premiums follows the process below:

● **Step1:** Count the temperature, precipitation, and number of natural disasters at the location of the historical landmark to be measured, $\mathbf{X}=\{\mathbf{T}, \mathbf{P}, \mathbf{W}\}$:

T is the average annual temperature data for the last 10 calendar years at the historic landmark site,

P is the precipitation data for the last 10 calendar years for the location of the historical landmark,

W is the number of extreme weather occurrences per year data for the last 10 calendar years at the historic landmark site,

and determine the corresponding regression coefficients and disturbance terms to find the corresponding regression equations:

$$Y = \beta_0 + \beta_1 * T + \beta_2 * P + \beta_3 * W$$

Taking the insurance loss percentage Y as the cost of coverage and assigning a certain percentage to different regions when facing them by examining their predominantly economic status factors.

The triple linear regression equation was fitted by MATLAB as follows:

$$Y = 4.5237 + 0.00593 * T + 0.000902 * P + 0.00098 * W$$

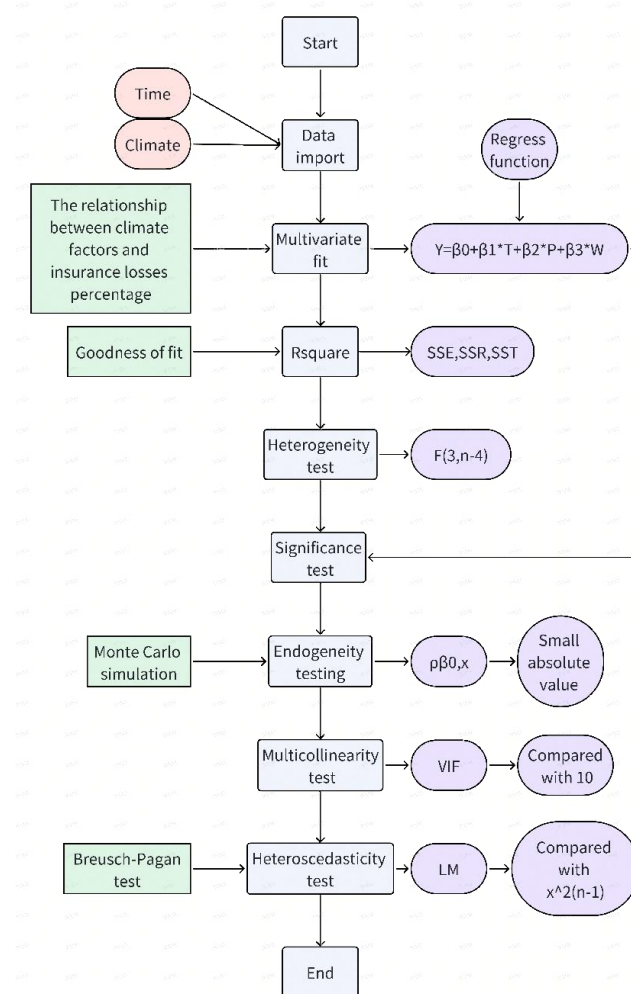


Figure 2:Flow Chart for Multiple Linear Regression

Calculate the corresponding data to make the following table:

Table 2.Value of the corresponding data

Source	SS	df
Model	0.0189	3
Residual	0.0247	6
Total	0.0436	9

Then calculate R^2_{adjusted} ,

$$\text{where } R^2_{\text{adjusted}} = 1 - [\text{SSE}/(n-k-1)]/[\text{SST}/(n-1)]$$

The reason for not using R^2 : as the number of independent variables increases, R^2 increases regardless of whether or not the introduced independent variables are actually correlated with the dependent variable, creating the illusion that the goodness of fit is better than it actually is, and for that reason, this is computed in this paper to avoid this problem. Calculated to obtain $R^2_{\text{adjusted}}=0.35$ the state of fit is favorable.

●**Step2:** Perform a heterogeneity test

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_1: \beta_1, \beta_2, \beta_3 \text{ are not all equal to } 0$$

The statistic to be tested is $F(3, n-4)$, and the acceptance of the original hypothesis is judged at the 95% confidence level. If the hypothesis is accepted, then all regression coefficients are insignificant and the constructed regression equation is faulty. If the original hypothesis is rejected, it means that not all regression coefficients are insignificant.

$$F(3,6)=4.76$$

Checking the table F distribution shows that the original hypothesis should be rejected at 95% confidence level and the regression coefficient is significant.

●**Step3:** Significance test on β_i

The standard errors of the regression coefficients and the perturbation terms are found, and the test statistic to be tested is t , and the corresponding P-value is found and compared with 0.05, which can be used to determine whether the regression coefficients and the perturbation terms are significant or not at the 95% confidence level. If $P < 0.05$, it means that β_i is significantly different from 0 at the 95% confidence level, and vice versa, it means that β_i is not significantly different from 0.

Table 3: Table made from the calculated corresponding values

Source	Coef	Std.Error	t	P> t	[95%.Conf.Internal]
T	0.00593	0.0472	0.126	0.126	1.8125
p	0.000902	0.0102	0.088	0.088	1.8125
W	0.00098	0.0046	0.213	0.213	1.8125
-cons	4.5237	12.70	0.356	0.356	1.8125

●Step4: Endogeneity test

Monte Carlo simulations were performed with multiple sets of data using the following three equations and plotted with the correlation coefficients on the horizontal axis and the corresponding regression coefficients on the vertical axis, respectively:

$$<1> y = \beta_1 * T + \beta_0 + b_1$$

$$\rho_{\beta_0, T} = 0.0084$$

$$<2> y = \beta_2 * P + \beta_0 + b_2$$

$$\rho_{\beta_0, P} = 0.00139$$

$$<3> y = \beta_3 * W + \beta_0 + b_3$$

$$\rho_{\beta_0, W} = 0.00074$$

All three are relatively small in absolute terms and are not endogenously severe.

●Step5: Tests for multicollinearity

$$VIF_m = \frac{1}{1 - R_{1-3 \setminus m}^2}$$

$$VIF = \max \{VIF_1, VIF_2, VIF_3\}$$

If $VIF > 10$, it means that the multicollinearity is serious and vice versa, it means that there is no serious multicollinearity problem.

Calculated from the data,

$$VIF = 2.1947 < 10,$$

it can be seen that the multicollinearity is not serious.

●Step6: BP test for heteroscedasticity

$$\varepsilon_i^2 = \delta_1 + \sum_{j=2}^n \delta_j * x_i + \text{error}_i$$

$$H_0: \delta_2 = \delta_3 = \delta_4 = 0$$

$$H_1: \delta_2, \delta_3, \delta_4 \text{ are not all equal to } 0$$

Construct the statistic $LM = nR^2 \sim \chi^2(n-1)$, find the corresponding p-value and test it at 95% confidence level. That is, if $P < 0.05$, the original hypothesis is rejected and there is significant heteroscedasticity, and conversely if $P > 0.05$, the original hypothesis is accepted and there is no significant heteroscedasticity.

Calculated from the data,

$$\delta_1 = 1.5572 * 10^{-14},$$

$$\delta_2 = -4.1139 * 10^{-16},$$

$$\delta_3 = 5.2390 * 10^{-18},$$

$$\delta_4 = -1.8693 * 10^{-17},$$

$$\chi^2(9) = 16.919,$$

$P > 0.05$, there is no significant heteroscedasticity.

●Step7: Calculation of premiums

$$Y = 4.5237 + 0.00593 * T + 0.000902 * P + 0.00098 * W,$$

$$\text{Insurance Premium} = Y * \text{Total premium},$$

Insurance Premium = $(4.5237 + 0.00593 * T + 0.000902 * P + 0.00098 * W) * \text{Total premium}$, where Y means the percentage of insured losses.

3.1.4 Analysis of the Result

The data collected resulted in a multivariate linear regression on the cost of insurance versus climate change and the occurrence of extreme weather in Yellowstone, Wyoming, U.S.A. The resulting regression equation passed all five tests and the adjusted goodness-of-fit showed a good state of fit. The relevance of this fitted equation is presented below:

$$\beta_1=0.006$$

$$\beta_2=0.0009$$

$$\beta_3=0.001$$

$$\beta_0=4.5$$

The approximated regression equation is as follows:

$$Y=4.5+0.006*T+0.0009*P+0.001*W$$

The relevance of the regression equation is:

For every 1 degree Fahrenheit increase in temperature, the average increase in insured losses is 0.006%;

For every 1mm increase in precipitation, the average increase in insured losses is 0.0009%;

For every 1 increase in the number of extreme weather events, the average increase in insured losses is 0.001%.

3.1.5 Verification of the actual application effect of the mode

In order to test whether the model we derived is generalizable and can effectively predict the cost of insurance in different regions, we conducted a single multiple regression on the Forbidden City of Shenyang, Liaoning, China, following the six steps described above, to test it and to examine whether the adjusted goodness-of-fit is good.

●**Step1:** Find the average annual temperature, precipitation, and the number of extreme weather occurrences in Shenyang Imperial Palace in the past 20 years, and find the ternary linear regression equation Y.

$$Y=4.3943+0.00581*T+0.000872*P+0.00096*W$$

$$R_{adjusted}^2=0.54$$

The model has good goodness of fit.

●**Step2:** Perform a heterogeneity test:

$$H_0:\beta_1=\beta_2=\beta_3=0$$

$$H_1: \beta_1, \beta_2, \beta_3 \text{ are not all equal to } 0$$

$$F(3,n-4)=F(3,6)=4.76$$

$$P<0.05$$

The original hypothesis is rejected and the regression coefficient is significant.

●**Step3:** Significance test of regression coefficients

Table 4. Significance test of regression coefficients

Source	Coef	Std.Error	t	P> t	[95%.Conf.Interval]
P	0.00581	0.0394	0.147	0.147	1.8125
T	0.000872	0.0198	0.044	0.044	1.8125
W	0.00096	0.0043	0.223	0.223	1.8125
-cons	4.3943	15.4	0.285	0.285	1.8125

●**Step4:** An endogeneity test is performed.

$$<1>y=\beta_1*T+\beta_0+b_1$$

$$\rho_{\beta_0,T}=0.0121$$

$$<2>y=\beta_2*P+\beta_0+b_2$$

$$\rho_{\beta_0,P}=0.00087$$

$$<3>y=\beta_3*W+\beta_0+b_3$$

$$\rho_{\beta_0,W}=0.00164$$

All three are relatively small in absolute terms and are not endogenously severe.

●**Step5:** Test for multicollinearity:

VIF=4.6872<10, indicating that the multicollinearity of the model is not serious.

●**Step6:** Heteroskedasticity test:

$$\varepsilon_i^2 = \delta_1 + \sum_{j=2}^n \delta_j * x_i + \text{error}_i$$

$$H_0: \delta_2 = \delta_3 = \delta_4 = 0$$

$$H_1: \delta_2, \delta_3, \delta_4 \text{ are not all equal to } 0$$

$$\delta_1 = 9.5742 * 10^{-15};$$

$$\delta_2 = -4.5563 * 10^{-16};$$

$$\delta_3 = 8.1961 * 10^{-17};$$

$$\delta_4 = 1.54533 * 10^{-16};$$

$$\chi^2(9) = 16.919$$

P<0.05, there is no significant heteroskedasticity.

●**Step7:** Verification gives rise to Insurance Premium=Y+θ.

In summary, the established ternary regression model for the Shenyang Imperial Palace corresponding insurance cost of good fit and passed the test. Therefore, the model in different areas, still has a good predictability, fit and actual situation, the future can be used for more different areas of insurance cost prediction analysis

3.1.6 Strength and Weakness

- Strength1:** The multivariate linear regression model allows for a quick and intuitive analysis of the linear relationship between the amount of insurance

payouts and the amount of precipitation, temperature, and the number of extreme weather occurrences.

- **Strength2:** Multiple linear regression can accurately measure the correlation between the variables, which is more accurate and close to the actual situation than univariate regression.
- **Strength3:** Multiple linear regression is easy to perform and yields unique results for the same set of data.
- **Weakness1:** The nonlinear relationship between precipitation, temperature, extremes, the number of weather occurrences and the dependent variable, and the interaction effects of the variables may have been neglected.
- **Weakness2:** It is only through speculation that we arrive at what kind of factor to choose and what kind of expression to use, which affects the diversity and unpredictability of the factors, making the accuracy of the regression analysis limited.

QUESTION 2

3.2 Recommendations on whether or not to Proceed with Real Estate

Development and Construction at a Particular Site

Predictions of future climatic conditions can be obtained by using one-way linear regression to predict future climatic conditions. The results of the data are taken into a multiple linear regression model and analyzed to determine the cost of insuring a site and, depending on the level of risk, whether or not a site should be developed.

QUESTION 3

3.3 Basic Model of Question 3

3.3.1 Terms, Definitions and Symbols

In this section, in order to facilitate the use of symbols and the reader's reading in subsequent articles, we will give the use of symbols in the model with their corresponding explanations. Some important mathematical notations used in this paper are listed in Table 5.

Table 5: Terms, definitions and symbols in basic model

Symbol	Definitions
T	Temperatures
P	Precipitation
W	Number of occurrences of extreme weather

ϵ_i^2 (i=1,2,3)	Error term
β_T	Regression coefficient of time on temperature
β_P	Regression coefficient of time on precipitation
β_W	Regression coefficient of time on the number of extreme weather events

3.2.2 Assumptions

Assumption1: A linear relationship is satisfied between the independent and dependent variables, respectively.

Justification: In the paper, the test of significance of regression coefficients was conducted and the goodness of fit was found to be significant and good, so the variables can be considered as linear.

Assumption2: The error term is a spherical perturbation term.

Justification: The Monte Carlo test for endogeneity and the BP test for heteroskedasticity are performed in the following sections, respectively, and the results show that the endogeneity is not significant and the error term is not significantly correlated with any of the independent variables, so that it can be regarded as a spherical perturbation term.

Assumption3: The error terms are normally distributed.

Justification: This error term is influenced by a variety of independent factors and is a random variable, while the distribution of a large number of independent random variables tends to be normal, so the error term should follow a normal distribution.

3.3.3 The Foundation of Model

In order for the multiple linear regression equations to predict the future cost of coverage for the Statue of Liberty in New York State, three one-way linear regression equations will be constructed to predict the future temperature, precipitation, and the number of extreme weather days in New York, respectively. This will be used as a basis for substituting into the multiple linear regression equations to predict the future cost of coverage.

Procedures: Establish a regression equation with time as the independent variable and temperature, precipitation, and days of extreme weather as the dependent variables. Then use the multiple linear regression in Question 1 to predict the cost of coverage and then the premium.

- **Step1:**Calculate the one-way linear regression equation:
T is the average annual temperature data for the last 10 calendar years at the location of the historical landmark;
P is the precipitation data for the location of the historical landmark for the last 10 calendar years;
W is the number of extreme weather occurrences per year for the last 10 calendar

years at the location of the historical landmark;

Three one-way linear regression equations with T, P, and W as dependent variables and time τ as independent variable are derived to predict the future climate conditions and the number of extreme weather occurrences.

The regression equation was found through MATLAB as follows:

$$T = -150 + 0.09940 * \text{time}$$

$$P = -13310 + 7.16303 * \text{time}$$

$$W = -13310 + 7.16303 * \text{time}$$

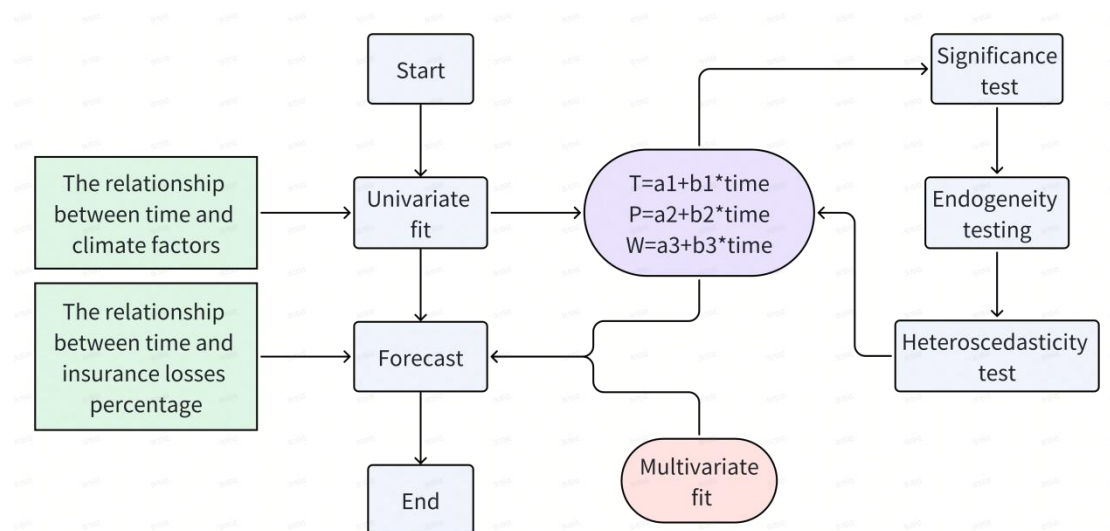


Figure 3: Flowchart for predicting the future by combining three one-dimensional linear fits

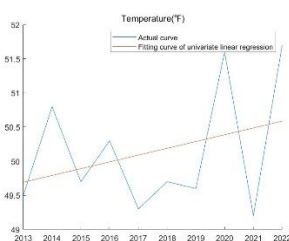


Figure 4: Regression Equation of T

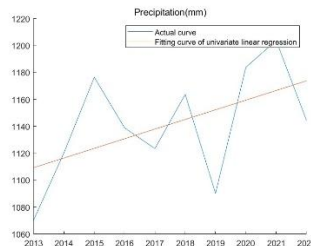


Figure 5: regression equation of P

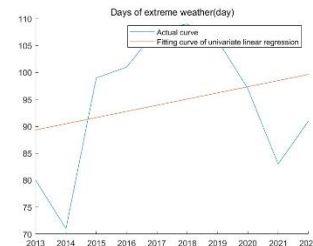


Figure 6: regression equation of W

● Step2: Significance test of regression coefficients:

The standard errors of the regression coefficients and the perturbation terms are found, and the test statistic to be tested is t, and the corresponding P-value is found and compared with 0.05, which can be used to determine whether the regression coefficients and the perturbation terms are significant or not at the 95% confidence level. If $P < 0.05$, it means that β_i is significantly different from 0 at 95% confidence level and vice versa.

The following table was obtained through MATLAB:

Table 6: Table with temperature as the dependent variable

Source	Coef	Std.Error	t	P> t	[95%.Conf.Interval]
t	0.09940	0.2857	0.3479	0.3479	1.8125
-cons	-150	664.5	-0.2257	0.2257	1.8125

Table 7: Table with precipitation as the dependent variable

Source	Coef	Std. Error	t	P> t	[95%.Conf.Interval]
t	7.16303	106.82	0.0671	0.0671	1.8125
-cons	-13310	69785.4	-0.1907	0.1907	1.8125

Table 8: Table with number of occurrences of extreme weather as the dependent variable

Source	Coef	Std. Error	t	P> t	[95%.Conf.Interval]
t	1.14545	5.8794	0.1948	0.1948	1.8125
-cons	-2216	4686.7	-0.4728	0.4728	1.8125

$P1 < 0.05, P2 < 0.05, P3 < 0.05$, the regression coefficients are all significantly different from 0.

● **Step3:** Endogeneity test:

$$<1> y = \beta_1 * T + \beta_0 + b_1$$

$$\rho_{\beta_0, T} = 0.0092$$

$$<2> y = \beta_2 * P + \beta_0 + b_2$$

$$\rho_{\beta_0, P} = 0.00054$$

$$<3> y = \beta_3 * W + \beta_0 + b_3$$

$$\rho_{\beta_0, W} = 0.00099$$

All three are relatively small in absolute terms and are not endogenously severe.

● **Step4:** Heteroscedasticity test:

$$\epsilon_i^2 = \delta_1 + \sum_{j=2}^{10} \delta_j * x_i + \text{error}_i, \text{ where } i=1,2,3;$$

$$H_0: \delta_2 = \delta_3 = \delta_4 = \dots = \delta_{10} = 0$$

$$H_1: \delta_2, \delta_3, \delta_4, \dots, \delta_{10} \text{ are not all equal to } 0$$

Separate the three sets of linear regressions corresponding to $\chi^2(9)$

$P1 < 0.05, P2 < 0.05, P3 < 0.05$, there is no significant heteroskedasticity.

● **Step5:** Calculate the percentage of insured losses

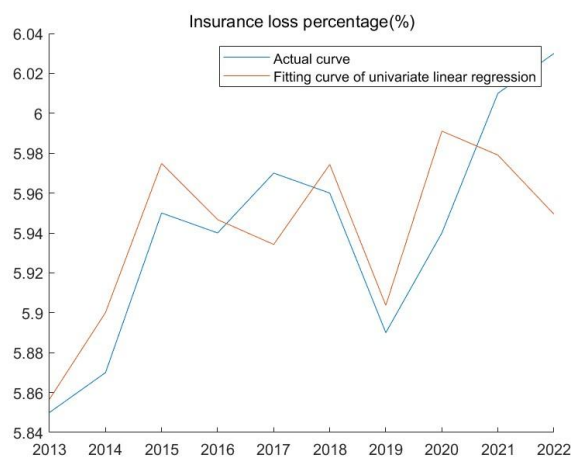


Figure 7: New York Insurance loss percentage

$$Y = 4.4945 + 0.0061 * T + 0.00091 * P + 0.0011 * W$$

$$T = -150 + 0.09940 * \text{time}$$

$$P = -13310 + 7.16303 * \text{time}$$

$$W = -2216 + 1.14545 * \text{time}$$

Substituting the T,P,W obtained from the univariate regression into the multivariate regression equation yields the percentage of insurance loss

$$Y = -10.97 + 0.008385 * \text{time},$$

and then Insurance Premium = $Y * \text{Total premium}$

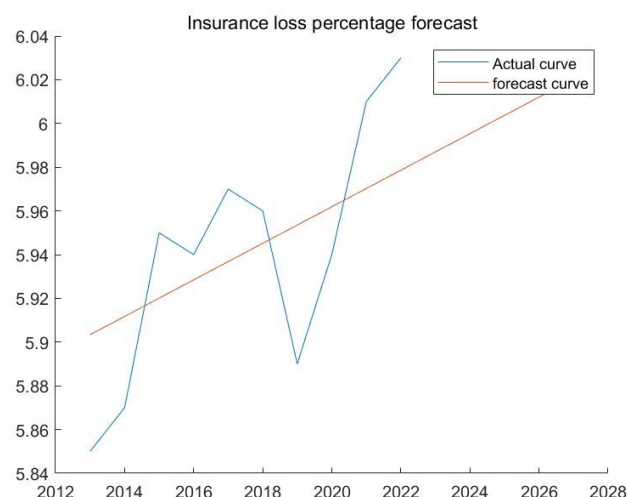


Figure 8: New York Insurance loss percentage forecast

3.3.4 Advice on Protecting Historic Landmarks

The regression analysis of climate change shows that there is an increasing trend in precipitation, the temperature is also increasing year by year, and the number of extreme weather days is also increasing, which correspondingly leads to an increase in the cost of safeguarding historical landmarks. In the case of China, from 1951 to 2021, the average annual surface temperature in China increased by 0.26°C per decade, and the average annual precipitation showed an increasing trend, with obvious regional differences in precipitation changes [1]. Worldwide, global warming has also caused more extreme and severe events [2]. While the climate is gradually moving out of the livable range, various properties, material or spiritual, inexpensive or expensive, common or scarce, are also at risk of destruction, in which the wear and tear and destruction of some historical landmarks that carry cultural values and spirituality have undoubtedly attracted a great deal of attention. Philosophical propositions such as the Ship of Theseus, they stand out for their uniqueness. If environmental and climatic conditions are allowed to continue to deteriorate, many historical landmarks may face high insurance costs while facing a higher risk of destruction by nature, and should therefore be protected in the following ways.

First, appropriate catastrophe insurance should be purchased for historic landmarks based on their value and the risk of disasters they face. Insurance can provide assistance for the reconstruction of a historical landmark after a disaster, greatly reducing the economic burden of the remodeling task, but at the same time, in the case of a historical landmark with good protection, uninterruptedly investing a large amount of money into it for a rainy day may result in a long-term economic expenditure, which may be questioned. However, as in the case of Notre Dame de

Paris, a world-renowned historic building, the cost of repairing it after the fire, which amounted to about €150 million, was too high to be insured. Therefore, it is also important to strengthen management and maintenance, and it is necessary to develop a sound system of use and visitation, pay attention to safety and security, and carry out regular maintenance to prevent permanent damage caused by long-term natural erosion. For important historical landmarks that cannot be protected in situ, consideration can be given to relocating them to other places for restoration; for those that are better preserved, such as the Statue of Liberty, in-situ preservation measures can be taken by designating a certain area as a unit for heritage protection, and restricting man-made activities within the area that may cause harm.

At the same time, from a long-term perspective, energy conservation and emission reduction, curbing global warming, stopping the further deterioration of the environment, and reversing the trend of an increase in the number of extreme weather events are the fundamental ways to safeguard historical landmarks from being devoured by nature. Human industrial civilization can tame nature, but inadvertently, nature's retaliation will destroy the spiritual wealth of mankind, which requires considerable attention.

3.3.5 Analysis of the Result

$$T = -150 + 0.09940 * \text{time},$$

indicating that for each year that passes, the regional temperature increases by an average of 0.09940°F.

$$P = -13310 + 7.16303 * \text{time},$$

indicating that for each year that passes, regional precipitation increases by an average of 7.16303 mm.

$$W = -2216 + 1.14545 * \text{time},$$

indicating that for each year that passes, the regional average increase in the number of days of severe weather occurs is 1.14545 days.

3.3.6 Strength and Weakness

- **Strength1:** Modeling is quick and easy, easy to operate, does not require very complex calculations, and still runs fast in the case of large amounts of data.
- **Strength2:** A linear relationship between time and each of the three dependent variables can be obtained from the coefficients, giving an understanding and interpretation of each variable.
- **Weakness:** Cannot fit nonlinear data well. So you need to determine if the variables are linearly related to each other first. The nonlinear relationship between the time of the independent variable and the dependent variable and the interaction effects of the variables may be overlooked.

IV. Conclusions

4.1 Conclusions of the problem

The problem can be divided into three main questions: first, to derive and demonstrate a model for determining whether or not insurance coverage should be provided under certain circumstances for two regions on different continents; second, to adapt the resulting insurance model to determine whether or not development should take place on certain sites; and third, to analyze a historical landmark using the resulting model and make recommendations on the cost of preserving it.

- For problem 1, we first analyzed Yellowstone Park in Wyoming to obtain a multiple linear regression equation with precipitation, temperature, and the number of extreme weather events as the independent variables and the percentage of insurance losses as the dependent variable. The equation was also subjected to a variety of tests for significance and validity. Then the feasibility of insuring Yellowstone Park was analyzed. Then we demonstrated and tested the applicability of the regression equation with data from the Shenyang Imperial Palace in Liaoning, and analyzed the feasibility of insuring the Shenyang Imperial Palace.
- For problem 2, we analyzed the impact of different factors on the model analysis, and for different regions, our model may be affected by other factors and adjusted accordingly to determine whether it is feasible to develop and build in a certain location.
- For problem 3, we used three sets of one-way linear regression to obtain the future precipitation, temperature, and the number of extreme weather events, and substituted the data into the multiple linear regression equations to estimate the percentage of insurance loss, and then made a judgment on whether to carry out the insurance or not, and gave relevant suggestions for the protection of historical landmarks.

4.2 Methods used in our models

4.2.1 *Multiple Linear Regression (MLR)*

Precipitation, temperature, and the number of extreme weather events are all considered to have an impact on the cost of insuring a historical surface, and all three factors are linearly related to the cost of insurance. Thus, multiple linear regressions were used to quickly and intuitively analyze the impact of these factors on insurance costs.

4.2.2 *Monte Carlo simulation*

For the analysis of model endogeneity, Monte Carlo simulation was used. This allows for a clearer picture than deterministic predictions based on a large amount of stochastic raw data, and a more intuitive analysis of the magnitude of endogeneity.

4.2.3 One-dimensional Linear Regression

Based on the conjecture that precipitation, temperature and the number of extreme weather occurrences increase year by year in a linear relationship over time, a one-way linear regression approach was adopted for their analysis. This makes it easy and quick to. Predict the future of these climatic conditions. It is also possible to further predict the future cost of insurance in a given location.

V. Future Work

5.1 Advantages of the model

- Through the multiple linear regression model, our easy operation can quickly and intuitively analyze the linear relationship of multiple variables with unique and accurate results. And after multiple tests, the significance and validity of the model are also guaranteed. The analysis of multiple factors is also more real and reliable, close to the actual situation.
- Monte Carlo simulation provides a wide range of possible outcomes and the probability of each outcome derived from a large sample of random data. It provides a clearer picture than deterministic forecasting.
- One-way linear regression modeling is convenient and easy to operate, does not require very complex calculations, and still runs fast in the case of large amounts of data. And it can intuitively give an explanation of how the independent variable changes with the dependent variable, and it can also have a corresponding explanation for the error term.

5.2 Disadvantages of the model

- Multiple linear regression models may ignore the nonlinear relationships between various independent and dependent variables and the interaction effects between variables. However, if too many variables are introduced, overfitting may be triggered. Moreover, the multiple linear regression analysis is only through speculation to conclude what kind of factors are chosen and what kind of expressions are used, which affects the diversity and unpredictability of the factors, and makes the accuracy of the regression analysis limited.
- Monte Carlo simulations require massive amounts of data to analyze the problem. Even for very simple problems, a large number of operations are required. This may complicate the problem. The amount of data given in this paper is not sufficient, which may result in inaccurate results.
- Univariate linear regression analysis does not fit nonlinear data well, so it is necessary to first determine whether the variables are linearly related to each other, which may ignore the nonlinear relationship between the time of the independent variable and the dependent variable, as well as the interaction effects of the variables.

5.3 Improved method of the model

- For the effect of temperature, precipitation and extreme weather days on insurance costs. A non-linear regression model can be attempted to analyze this, and the applicability of the model can be verified by determining whether the goodness of fit is good or not. And stepwise regression analysis can be used to determine the number of independent variables that have a significant effect on the dependent variable. The regression coefficients of the multiple regression model can also be analyzed and tested with more data to ensure its generalizability.
- In Monte Carlo simulations, one should aim to find to obtain more data and substitute it to ensure the accuracy of their results.
- For the analysis of the trend of temperature, precipitation and the number of extreme weather days over time, it is possible to make graphs, observe the trend from the graphs, and then use linear and non-linear equations to fit and check the goodness-of-fit, so that the trend of these climatic conditions over time can be analyzed more accurately.

VI. Letter of Recommendation on the Preservation of the Statue of Liberty

Located in New York State, the Statue of Liberty has been around for over 100 years now. It is not only a landmark of New York City, but also a symbol of the spirit of freedom, independence and democracy. As a New York landmark and a common heritage around the world, the future of the Statue of Liberty is in the spotlight. With the passage of time and the deterioration of climatic conditions, the frequent occurrence of extreme weather, the conservation of the Statue of Liberty is facing new challenges, and the issue has attracted extensive attention from the world. This is a manifestation of the importance and urgency of preserving the cultural heritage of mankind. After our team's analysis of the insurance and protection of the Statue of Liberty, we would like to give the following suggestions:

First, through the analysis of the model established by our team, we know that in the current situation, there is a trend of more severe weather conditions and the number of extreme weather occurrences in New York. In order to control the frequent occurrence of extreme weather, timely improvement of the environment, energy saving and emission reduction are the long-term measures for the protection of the Statue of Liberty.

Secondly, our team's analysis shows that the insurance risk of the Statue of Liberty is within acceptable limits and low. Insurance companies should agree to insure the Statue of Liberty against catastrophe at a lower price. Moreover, the Statue of Liberty's historical and cultural significance is of great value, and in the event of a major natural disaster, there is the potential for a large loss. Therefore, it is not very difficult to find an insurance company that is willing to cover the Statue of Liberty, and if we do not choose to insure the Statue of Liberty, the loss in case of a natural

disaster will be greater, so our team believes that it is feasible to purchase a suitable catastrophe insurance policy for the Statue of Liberty, which is necessary at the same time.

Thirdly, an area near the Statue of Liberty should be designated as an area for heritage protection. Within the area, restrictions should be imposed on man-made measures that may cause damage to the Statue of Liberty, such as discarding unextinguished cigarette butts, and certain punitive measures should be taken against those who violate the rules. Volunteers can be arranged to coordinate the management of the area to safeguard the Statue of Liberty from human damage.

Fourthly, regular maintenance of the Statue of Liberty can be arranged to prevent permanent damage caused by long-term natural erosion, while short-term damage can be repaired in a timely manner, reducing maintenance costs.

VII. References

- [1]China's Blue Book on Climate Change (2022) Released, Global Warming Trend Continues Many of China's Climate Change Indicators Break Observation Records in 2021. China Meteorological News, August,2022.
- [2]Global Warming and Climate Change; Investigators from Stanford University Have Reported New Data on Global Warming and Climate Change (Quantifying the influence of global warming on unprecedented extreme climate events)[J].The Business of Global Warming,2017,65-.

VIII. Appendix

```
clc;clear;close all;
```

```
% Data import
```

```
% Time
```

```
T = [];
```

```
% New York State data
```

```
x1 = []; % Precipitation
```

```
x2 = []; % Temperature
```

```
x3 = []; % Days of extreme weather
```

```
Y = []; % Insurance loss percentage
```

```
% Multivariate fitting of insurance premiums to climate elements
```

```
first = ones(length(T),1);
```

```
X0 = [first , x1, x2, x3];
```

```
[b,bint,r,rint,stats] = regress(Y,X0,0.05);
```

```
Fit0 = b(1) + b(2) .* x1 + b(3) .* x2 + b(4) .* x3 ;
```

```
figure(1);
```

```
hold on;
```

```
plot(T,Y);
```



```

X13 = [first , x1, x3];
[b13,bint,r,rint,stats] = regress(Y,X13,0.05);
Fit13 = b13(1) + b13(2) .* x1 + b13(3) .* x3;
Rsquare13 = 1 - sum( (Fit13 - Y).^2 ) ./ sum( (Y - mean(Y)).^2 );

X23 = [first , x2, x3];
[b23,bint,r,rint,stats] = regress(Y,X23,0.05);
Fit23 = b23(1) + b23(2) .* x2 + b23(3) .* x3;
Rsquare23 = 1 - sum( (Fit23 - Y).^2 ) ./ sum( (Y - mean(Y)).^2 );

Rsquaremax = max(max(Rsquare12,Rsquare13),Rsquare23);
VIFm = 1./(1-Rsquaremax);
VIFmstr = num2str(VIFm);
disp(['VIFm: ',VIFmstr])
if VIFm>10
    disp('Multilinearity is severe!')
else
    disp('Multilinearity is not severe!')
end;

% Heteroscedasticity test
disp('Heteroscedasticity test:')

residual = Y - b(1) - b(2) .* x1 - b(3) .* x2 - b(4) .* x3;
[bres,bint,r,rint,stats] = regress(residual,X0,0.05);
Fittres = bres(1) + bres(2) .* x1 + bres(3) .* x2 + bres(4) .* x3 ;
disp('residual = bres(1) + bres(2) .* x1 + bres(3) .* x2 + bres(4) .* x3, The parameters
are as follows:')
disp(bres(1))
disp(bres(2))
disp(bres(3))
disp(bres(4))

% Draw images with time as a variable for linear regression to predict climate change
XT = [first, T];

[bt1,bint,r,rint,stats] = regress(x1,XT,0.05);
Fitt1 = bt1(1) + bt1(2) .* T ;
figure(2);
hold on;
plot(T,x1);
plot(T,Fitt1);
legend('Actual curve','Fitting curve of univariate linear regression')
title('Precipitation(mm)')

```

```
[bt2,bint,r,rint,stats] = regress(x2,XT,0.05);  
Fitt2 = bt2(1) + bt2(2) .* T ;  
figure(3);  
hold on;  
plot(T,x2);  
plot(T,Fitt2);  
legend('Actual curve','Fitting curve of univariate linear regression')  
title('Temperature(°F)')
```

```
[bt3,bint,r,rint,stats] = regress(x3,XT,0.05);  
Fitt3 = bt3(1) + bt3(2) .* T ;  
figure(4);  
hold on;  
plot(T,x3);  
plot(T,Fitt3);  
legend('Actual curve','Fitting curve of univariate linear regression')  
title('Days of extreme weather(day)')
```

```
% Based on the expected climate change, obtain the expected percentage change in  
insurance losses  
year = 2013 : 1 : 2027;  
loss = b(1) + b(2) .* ( bt1(1) + bt1(2) .* year) + b(3) .* (bt2(1) + bt2(2) .* year) +  
b(4) .* ( bt3(1) + bt3(2) .* year) ;  
figure(5);  
hold on;  
plot(T, Y)  
plot(year, loss)  
legend('Actual curve','forecast curve')  
title('Insurance loss percentage forecast')
```